Cyber Security : Hardware and Storage

Day 2

Task 1

RAID	Min Disks	Redundancy	Speed	Use Case
RAID 0	2	None. One disk fails and everything is lost	Very fast reads and writes	Video editing scratch disks, temporary render or cache storage where data can be recreated
RAID 1	2	Full copy on each disk, survives 1 disk failure	Good reads, writes about the same as a single disk	Boot drives, small databases, WhatsApp-style chat backups
RAID 5	3	Survives 1 disk failure (parity rebuilds lost data)	Fast reads, slower writes (parity calculation)	File servers, YouTube-style video archives where capacity matters
RAID 6	4	Survives 1 disk per mirror pair (often 2 total)	Very fast reads and writes	Busy databases, Instagram-style photo metadata and transaction systems

Task 2

Say you store WhatsApp chat backups on **Disk A** and **Disk B**. Every time a backup is written, the controller writes the identical data to both disks at the same time, so they are exact mirrors.

If Disk A dies, the array keeps running in a degraded state. Reads and writes continue from Disk B, so no backup is lost and users notice nothing. You replace Disk A with a new disk, and the RAID software copies everything

from Disk B onto it (a rebuild). Until the rebuild finishes you have no redundancy, so a second failure would lose data.

RAID is not a backup. If you accidentally delete a file or ransomware encrypts it, both disks lose it at once.

Task 5

- **Capacity vs cost:** RAID 5 loses only one disk's worth of space to parity, so it's cheaper per usable TB. RAID 10 loses 50%.
- **Performance:** RAID 10 handles many small, random reads and writes better, with no parity overhead. RAID 5 has slow writes, which hurts busy workloads.
- **Failure and rebuild risk:** RAID 10 rebuilds quickly by copying one mirror. RAID 5 rebuilds slowly on large disks and is vulnerable to a second failure during that time. Large cloud providers often go beyond both, using erasure coding or replication across servers.